

Mohit

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Professional Summary:

AI and Machine Learning enthusiast with hands-on experience developing and deploying deep learning models in Computer Vision and NLP. Built a portfolio of high-accuracy models for medical diagnostics and real-time facial expression recognition using PyTorch and TensorFlow. Eager to contribute to innovative AI solutions in a challenging internship role.

Education Qualifications:

B.Tech in Artificial Intelligence and Data Science

Arya College of Engineering and IT, Rajasthan Technical University | CGPA: 8.8 (till 3rd semester) Expected Graduation: 2027

Skills:

- Programming Languages: Python, C++, SQL
- Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
- Natural Language Processing: NLTK, spaCy, Hugging Face Transformers, Text Preprocessing
- Computer Vision: OpenCV, CNNs
- Web Development: Flask, HTML5, CSS, JavaScript
- Developer Tools & Platforms: Git, GitHub, Docker, VS Code, Google Colab, Jupyter Notebook
- Core CS Concepts: Data Structures & Algorithms, Deep Learning, OOP, DBMS, Operating Systems, Computer Networks

Projects:

MINDAEYE | AI Medical Diagnosis Platform

[Github](#)

- Architected two **PyTorch deep learning** models from scratch, achieving 98% accuracy for brain tumor detection (4 classes) and 84% for diabetic retinopathy classification (5 stages).
- Developed the complete **Flask backend** to serve the AI models, manage medical image uploads, and dynamically generate downloadable PDF diagnostic reports for patients.
- Integrated a **Gemini API-powered chatbot** to provide users with real-time, interactive health assistance and automate responses to complex medical queries.
- **Tech Stack:** Python, PyTorch, Flask, Gemini API, OpenCV, Pandas, NumPy, FPDF, HTML/CSS, JavaScript

Rockfall | Predictive Safety System

[Website](#)

[Github](#)

- Designed and implemented a **Bidirectional LSTM with Attention model in PyTorch** to analyze 10-day sequences of **NASA weather and geospatial data**, proactively forecasting rockfall risk in open-pit mines.
- Developed a full end-to-end system, deploying the trained model via a **Flask REST API** to power an intuitive user dashboard for risk visualization and high-risk event alerts.
- Constructed a complete data processing pipeline using **Pandas and Scikit-learn to ingest**, clean, enrich, and transform raw NASA catalog data into feature-rich time-series sequences for model training.
- **Tech Stack:** Python, PyTorch, Flask, Scikit-learn, Pandas, NumPy, Jupyter Notebook

Emotion Detection | Real-Time Facial Expression Recognition

[Github](#)

- Fine-tuned a **pre-trained ConvNeXt Tiny model in PyTorch** to classify 7 core facial emotions (e.g., Happy, Sad, Angry), achieving high accuracy on benchmark datasets.
- Implemented a real-time pipeline using **OpenCV and dlib** to first detect faces in a live webcam feed and then classify their emotions instantly.
- **Tech:** Python, PyTorch, OpenCV, dlib, torchvision

Achievements:

- Finalist (Top 6 of 750+ teams) at Microsoft CodeForge 2025 for developing an AI healthcare platform using Flask, EfficientNet, and REST APIs, achieving 85%+ accuracy in medical image classification. [Link](#)
- Attendee, E-Summit 2025 (IIT Bombay) – Engaged in workshops and keynote sessions on entrepreneurship, innovation, and emerging technologies. [Link](#)
- 4th Place out of 78 teams at Arya Group of Colleges Hackathon, awarded ₹2000 cash prize for building an innovative AI-based healthcare solution, showcasing expertise in AI/ML and solution development. [Link](#)

Experience:

GDG ACEIT – Jaipur, Rajasthan, India

[Link](#)

Data Engineer Lead | Oct 2024 – Present

- Led AI/ML data workflows utilizing Python and SQL.
- Mentored peers in data engineering principles and machine learning concepts.

Certifications:

International Astronomical Search Collaboration (IASC) – Worked with NASA & Pan-STARRS on asteroid detection. [Link](#)

Kaggle – Completed certifications in Intro to Programming, Python, and Machine Learning. [Link](#)

NPTEL- Elite Certificate in Fundamentals of Object Oriented Programming (IIT Roorkee, Score:69%) [Link](#)